

Investment and Finance

Additional Notes for Continuous Time

December 2024

This document contains derivations related to certain parts of the slides, including those that were written on the board.

Stochastic calculus in continuous time

We have a process X_t , we know that it starts at a value of X_0 and how it evolves, and we are interested in how it will look like at time T , i.e., in X_T .¹ We want to use the definition:

$$X_T - X_0 = \int_0^T dX_t \quad (1)$$

i.e., the overall change between time 0 and T is the sum of small changes during the path.

If changes in X_t are deterministic, i.e., do not depend on any randomness, we can rewrite things as what you might be better used to. E.g.,

$$\frac{dX_t}{dt} = 2$$

is equivalent to writing

$$dX_t = 2dt \quad (2)$$

(basically we just “multiply” by the dt term) and we can write

$$X_T - X_0 = \int_0^T dX_t = \int_0^T 2dt = 2 \int_0^T dt = 2T.$$

Hence, $X_T = X_0 + 2T$. If you differentiate that, we indeed get (2).

When there is some randomness involved, we want to do the same as in (1), but of course things will have a slightly different interpretation. We have introduced randomness in a particular way, via Wiener processes z_t whose increments over a tiny time

¹The index t refers to time. Many people would actually write it as $X(t)$, so that it is more straightforward that we are talking about a function of time, as in $f(x)$. In finance people tend to write it as a subscript because otherwise formulas would look much bigger when printed.

interval dt , denoted by dz_t , are independently distributed: they are normal with expectation zero and variance dt . If you add multiple of them, then the sum of them will also be normal, the expectation will be the sum of the expectations, i.e., zero, and the variance will be the sum of the variances, which from $\int_0^T dt = T$ will be simply the time that passed by, T . So, we can write

$$\int_0^T dz_t = z_T - z_0,$$

with $z_T - z_0 \sim \mathcal{N}(0, T)$. If we set $z_0 = 0$ then $z_T \sim \mathcal{N}(0, T)$.

Now, combining the two above, if we have a starting point X_0 and $dX_t = a dt + b dz_t$, where a and b are constants, then

$$\begin{aligned} X_T - X_0 &= \int_0^T dX_t = \int_0^T (a dt + b dz_t) = \int_0^T (a dt + b dz_t) \\ &= \int_0^T a dt + \int_0^T b dz_t = a \int_0^T dt + b \int_0^T dz_t \\ &= aT + bz_T \end{aligned} \tag{3}$$

That is, $X_T = (X_0 + aT) + bz_T$. Since $z_T \sim \mathcal{N}(0, T)$, we get that $X_T \sim \mathcal{N}(X_0 + aT, b^2 T)$.

But what if the coefficients of dt and dz_t , i.e., a and b , are not constant? If they are deterministic (do not depend on any randomness), then we are still fine, because we can already integrate functions, just need to be a bit careful with the random part. So we would just write

$$\begin{aligned} X_T - X_0 &= \int_0^T dX_t = \int_0^T (a_t dt + b_t dz_t) = \int_0^T (a_t dt + b_t dz_t) \\ &= \int_0^T a(t) dt + \int_0^T b(t) dz_t. \end{aligned}$$

Since here all dz_t are independent normal with expectation zero and variance dt , we have that the expectation of that integral term is also zero, and the variance of one $b(t)dz_t$ term is $b^2(t)dt$, so the overall variance is $\int_0^T b(t)dt$. Therefore, X_T is normal with mean $X_0 + \int_0^T a(t)dt$ and with variance $\int_0^T b(t)dt$. But this is just for your information; you won't have to do anything like this in this course.²

So what if $a(t)$ and $b(t)$ are not deterministic? Then we need a trick: we need to transform our process into something that has deterministic drift and diffusion so we can work with them. The reason for this is that if we write, for example,

$$dS_t = \mu S_t dt + \sigma S_t dz_t,$$

then we cannot simply integrate the $\sigma S_t dz_t$ terms: today's z_t affects the change dS_t , which affects all future values of S_t , and hence all future random increments $\sigma S_t dz_t$.

²Notice that I rewrite a and b as functions in that equation, just to be explicit and have a notation closer to what you might know better.

Hence, we cannot say that all random increments are independent of each other, and we cannot add up their variances. Moreover, they would affect all future $\mu S_t dt$ terms too, so we cannot just integrate those as we did with a deterministic a_t .

Put differently: if you want to write down (continuously compounded) returns between dates t and $t + dt$, and between $t + dt$ and $t + 2dt$, i.e., over the next two small time intervals, we would need to calculate

$$\frac{S_{t+dt} - S_t}{S_t} + \frac{S_{t+2dt} - S_{t+dt}}{S_{t+dt}},$$

but this cannot be simplified easily, to write $\frac{S_{t+2dt} - S_t}{S_t}$, because the two denominators are slightly different.

Itô's Lemma

The trick is called Itô's Lemma, which says that for a smooth function $f(t, x)$ of two variables (i.e., one that can be differentiated as many times as we want) and a diffusion process X_t ,

$$dX_t = a_t dt + b_t dz_t,$$

where a_t and b_t do NOT need to be deterministic functions, but can depend on t and X_t , the process $f(t, X_t)$ satisfies

$$d(f(t, X_t)) = \left[\frac{\partial f}{\partial t}(t, x) + \frac{\partial f}{\partial x}(t, x)a_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, x) (b_t)^2 \right] dt + \frac{\partial f}{\partial x}(t, x)b_t dz_t.$$

Applying Itô's Lemma to a Geometric Brownian motion to obtain a closed-form expression for S_T

Suppose we write

$$\frac{dS_t}{S_t} = \mu dt + \sigma dz_t,$$

which is equivalent to

$$dS_t = \mu S_t dt + \sigma S_t dz_t. \tag{4}$$

We want to get the value S_T if S follows this process and at time 0 it has a value of S_0 .

Let us now consider the function $f(t, x) = \ln x$. I.e., it will not depend on time, just the underlying process. It is easy to check that

$$\frac{\partial f}{\partial t}(t, x) = 0, \quad \frac{\partial f}{\partial x}(t, x) = \frac{1}{x}, \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2}(t, x) = -\frac{1}{x^2}.$$

So, if we want to learn how the process $\ln S_t$ evolves over time, we need to use Itô's Lemma, together with $a_t = \mu S_t$ and $b_t = \sigma S_t$. We obtain the following:

$$d(\ln S_t) = \left[0 + \frac{1}{S_t} \mu S_t - \frac{1}{2} \frac{1}{S_t^2} (\sigma S_t)^2 \right] dt + \frac{1}{S_t} \sigma S_t dz_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dz_t.$$

But this is good, because we have now a process whose drift and diffusion are constant, so we can revisit (3):

$$\ln S_T - \ln S_0 = \int_0^T d(\ln S_t) = \int_0^T \left[\left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dz_t \right] = \left(\mu - \frac{1}{2}\sigma^2 \right) T + \sigma z_T. \quad (5)$$

That is, $\ln S_T = \ln S_0 + \left(\mu - \frac{1}{2}\sigma^2 \right) T + \sigma z_T$, where $z_T \sim \mathcal{N}(0, T)$. Taking the exponential of both sides, we end up with $S_T = S_0 e^{(\mu - \frac{1}{2}\sigma^2)T + \sigma z_T}$.

Deriving the BSM formulas with risk-neutral pricing

BSM formula or a European call

For simplicity, I perform the expectation calculations for $t = 0$, which simplifies all formulas by replacing $T - t$ by T . The general formulas should follow immediately. Also, the Wiener process is denoted by z .

Under the risk-neutral probability measure $\mathbb{Q}$,

$$\frac{dS_t}{S_t} = r dt + \sigma dz_t \iff S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma z_T}$$

(see above, by replacing μ with r), so the price of a European call option is

$$\begin{aligned} c_0 &= e^{-rT} \mathbb{E}^{\mathbb{Q}} [S_T - K]^+ \\ &= e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\left[S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma z_T} - K \right]^+ \right] \\ &= e^{-rT} \int_{-\infty}^{\infty} \left[S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \sqrt{T}z} - K \right]^+ n(z) dz \\ &= (*), \end{aligned}$$

with $n(z)$ being the pdf of a standard normal distribution. Now,

$$\begin{aligned} S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \sqrt{T}z} - K &\geq 0 \\ \iff z &\geq \frac{1}{\sigma \sqrt{T}} \left(\ln \left(\frac{K}{S_0} \right) - \left(r - \frac{1}{2}\sigma^2 \right) T \right) \equiv -d_2 \end{aligned}$$

and we can write

$$\begin{aligned} (*) &= e^{-rT} \int_{-d_2}^{\infty} \left(S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma \sqrt{T}z} - K \right) n(z) dz \\ &= e^{-rT} \left(S_0 e^{(r - \frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} e^{\sigma \sqrt{T}z} n(z) dz - K \int_{-d_2}^{\infty} n(z) dz \right) \\ &= e^{-rT} \left(S_0 e^{(r - \frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} e^{\sigma \sqrt{T}z} n(z) dz - K(1 - N(-d_2)) \right) \end{aligned}$$

$$\begin{aligned}
&= e^{-rT} \left(S_0 e^{(r-\frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} e^{\sigma\sqrt{T}z} n(z) dz - KN(d_2) \right) \\
&= (\square).
\end{aligned}$$

A direct calculation shows that

$$e^{\sigma\sqrt{T}z} n(z) = e^{\sigma\sqrt{T}z} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}((z-\sigma\sqrt{T})^2 - \sigma^2 T)}.$$

Using this last equality,

$$\begin{aligned}
(\square) &= e^{-rT} \left(S_0 e^{(r-\frac{1}{2}\sigma^2)T} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}((z-\sigma\sqrt{T})^2 - \sigma^2 T)} dz - KN(d_2) \right) \\
&= e^{-rT} \left(S_0 e^{rT} \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-\sigma\sqrt{T})^2} dz - KN(d_2) \right) \\
&= e^{-rT} \left(S_0 e^{rT} \int_{-d_2-\sigma\sqrt{T}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz - KN(d_2) \right) \\
&= e^{-rT} \left(S_0 e^{rT} \left(1 - N(-d_2 - \sigma\sqrt{T}) \right) - KN(d_2) \right) \\
&= e^{-rT} \left(S_0 e^{rT} N(d_2 + \sigma\sqrt{T}) - KN(d_2) \right) \\
&= e^{-rT} (S_0 e^{rT} N(d_1) - KN(d_2)),
\end{aligned}$$

where we implicitly defined $d_1 \equiv d_2 + \sigma\sqrt{T}$ for the last equality. Summing up,

$$c_0 = S_0 N(d_1) - e^{-rT} KN(d_2),$$

with

$$\begin{aligned}
d_1 &\equiv \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} \\
d_2 &\equiv \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.
\end{aligned}$$

This is the Black-Scholes-Merton formula for a European call option.

Δ of a European call

Inspection shows that the BSM formula for a European call is homogeneous of degree one in the underlying price and the strike price. In mathematical terms, it means that

$$c_0(k \times S_0, k \times K, r, \sigma, T) = k \times c_0(S_0, K, r, \sigma, T).$$

for any $k \geq 0$. But then, thanks to Euler's Homogeneous Function Theorem, we can write

$$c_0(S_0, K, r, \sigma, T) = S_0 \frac{\partial c_0}{\partial S_0}(S_0, K, r, \sigma, T) + K \frac{\partial c_0}{\partial K}(S_0, K, r, \sigma, T).$$

Comparing this last expression to the BSM formula, we can deduce that

$$\Delta_{\text{call}} = \frac{\partial c_0}{\partial S_0}(S_0, K, r, \sigma, T) = N(d_1).$$

Alternatively, you could just do the differentiation carefully.

BSM formula for a European put

Combining the put-call parity and the BSM formula for a European call yields

$$\begin{aligned} p_0 &= c_0 - S_0 + e^{-rT}K \\ &= -S_0(1 - N(d_1)) + e^{-rT}K(1 - N(d_2)) \\ &= e^{-rT}KN(-d_2) - S_0N(-d_1), \end{aligned}$$

which is the Black-Scholes-Merton formula for a European put option.

Δ of a European put

Recalling the put-call parity, the definition of the Δ of a derivative in the BSM setting, and the expression for the Δ of a European call yields

$$\begin{aligned} c_0 - p_0 &= S_0 - e^{-rT}K \\ \xRightarrow{\partial_{S_0}(\cdot)} \partial_{S_0}(c_0) - \partial_{S_0}(p_0) &= 1 \\ \iff \Delta_{\text{put}} &= \Delta_{\text{call}} - 1 = N(d_1) - 1. \end{aligned}$$